

Question 1 (10 Marks)**Marks**

- (a) Factorise fully $4x^3 - 500$. 2
- (b) Convert $0.4\dot{7}\dot{6}$ to a fraction in its simplest terms. 2
- (c) Simplify $\frac{6x-4y}{3x-2y}$. 1
- (d) Solve $|2x-3| > 7$ and graph your solution on the number line. 2
- (e) Express $\frac{5}{\sqrt{3}-1} - \frac{2}{2+\sqrt{3}}$ with a rational denominator. (3)

Question 2 (10 Marks) **START A NEW PAGE**

- (a) A circle has the equation $x^2 - 14x + y^2 + 4y + 49 = 0$
- (i) Find the co-ordinates of the centre and the length of the radius. 3
- (ii) Hence or otherwise, sketch the circle showing all essential features including intercepts. 1
- (b) Sketch the following showing all the essential features including intercepts and asymptotes.
- (i) $y = (x-1)^2 + 3$ 2
- (ii) $y = \frac{1}{2x-1} - 2$ 2
- (c) State the domain and range of $y = \frac{1}{\sqrt{2x+9}}$. 2

Question 3 (10 Marks) **START A NEW PAGE**

- (a) Solve $\frac{x}{x-3} \leq 2$. 3
- (b) A rectangle has a perimeter of 40 centimetres and a length of x centimetres.
- (i) Show that the area is given by $A = 20x - x^2$. 1
- (ii) Find the length of the rectangle when the area is 96 cm^2 . 1

Question 3 continued Page 2.

Question 3 Continued**Marks**

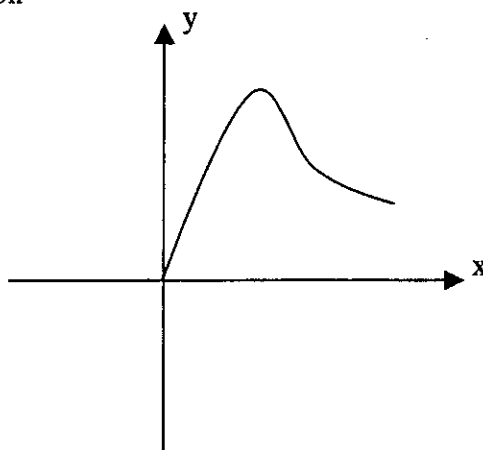
(c) Solve $|x + 1| = 2 + 3x$

3(d) The sketch of $f(x)$ is shown belowUsing separate diagrams copy and complete the sketch to make $y = f(x)$

(i) An odd function

1

(ii) An even function

1**Question 4** (10 Marks) **START A NEW PAGE**(a) Shade the region where $y \leq \sqrt{9 - x^2}$ and $y > 3x$ hold simultaneously.**3**(b) Given $f(x) = 3(2^x) - 2(3^x)$, prove that $f(x+1) = 6(2^x - 3^x)$ **2**(c) Given $f(x) = \frac{2x^2}{x^2 - 16}$

(i) Find the vertical asymptotes.

1

(ii) Find the horizontal asymptotes.

1(iii) Is the $f(x)$ odd, even or neither? Justify your answer.**1**

(iv) Sketch the curve, showing all essential details including intercepts and asymptotes.

2**END OF PAPER**

Question 1

$$a^3 \pm b^3$$

$$\begin{aligned}
 \text{a) } 4x^3 - 500 &= 4(x^3 - 125) = (a \pm b)(a^2 \mp ab + b^2) \\
 &= 4(x^3 - 5^3) \\
 &= 4(x-5)(x^2 + 5x + 25)
 \end{aligned}$$

b) Let

$$x = 0.\dot{4}7\dot{6}$$

$$\textcircled{x} = 0.4767676\ldots \quad \text{--- ①}$$

$$1000x = 476.7676\ldots \quad \text{--- ②}$$

$$10x = 4.7676\ldots \quad \text{--- ③}$$

$$\textcircled{2} - \textcircled{3} \quad 990x = 472$$

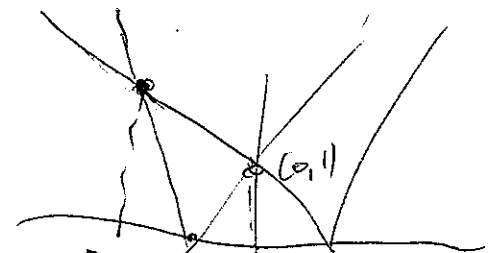
$$x = \frac{472}{990} \quad \checkmark$$

$$x = \frac{236}{495}$$

$$\therefore 0.\dot{4}7\dot{6} = \frac{236}{495} \quad \checkmark$$

$$\begin{aligned}
 1-x &= |2x+1| \quad x=0 \\
 2x+1 &= 1-x \\
 2x+1 &= x-1 \quad x=-2
 \end{aligned}$$

$$|x-1| \leq |2x+1|$$



$$x \leq -2, \quad x \geq 0$$

$$\begin{aligned}
 \text{c) } \frac{6x-4y}{3x-2y} &= \frac{2(3x-2y)}{(3x-2y)} \\
 &= 2
 \end{aligned}$$

d) $|2x-3| > 7$

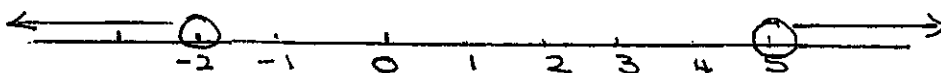
$$2x-3 < -7 \quad \text{or} \quad 2x-3 > 7$$

$$2x < -4$$

$$2x > 10$$

$$x < -2$$

$$x > 5$$



$$e) \frac{5}{\sqrt{3}-1} - \frac{2}{2+\sqrt{3}}$$

$$\frac{5}{\sqrt{3}-1} = \frac{5}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1}$$

$$= \frac{5(\sqrt{3}+1)}{2}$$

$$\frac{2}{2+\sqrt{3}} = \frac{2}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}}$$

$$= 2(2-\sqrt{3})$$

$$\therefore \frac{5}{\sqrt{3}-1} - \frac{2}{2+\sqrt{3}} = \frac{5(\sqrt{3}+1) - 4(2-\sqrt{3})}{2}$$

$$= \frac{5\sqrt{3}+5-8+4\sqrt{3}}{2}$$

$$= \frac{9\sqrt{3}-3}{2}$$

Question 2

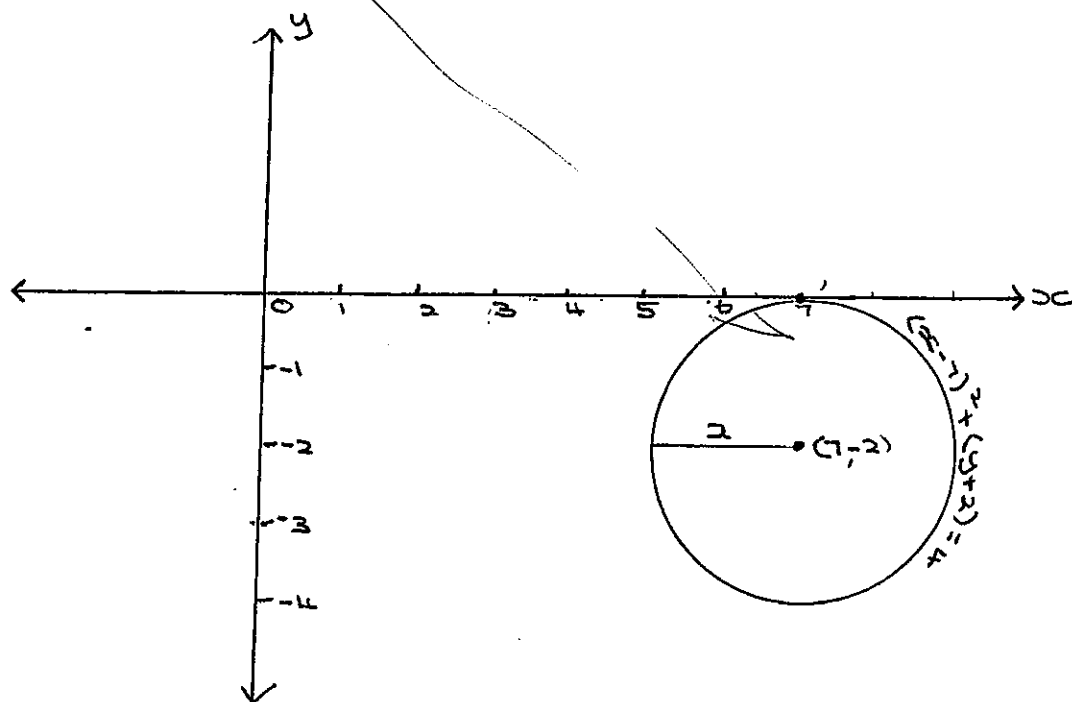
$$a) x^2 - 14x + y^2 + 4y + 49 = 0$$

$$x^2 - 14x + 49 + y^2 + 4y + 4 = 4$$

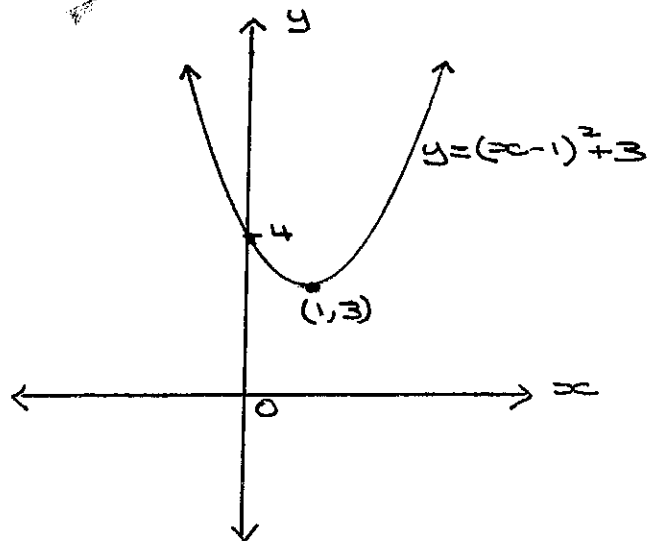
$$(x-7)^2 + (y+2)^2 = 2^2$$

$\therefore$ The centre is $(7, -2)$

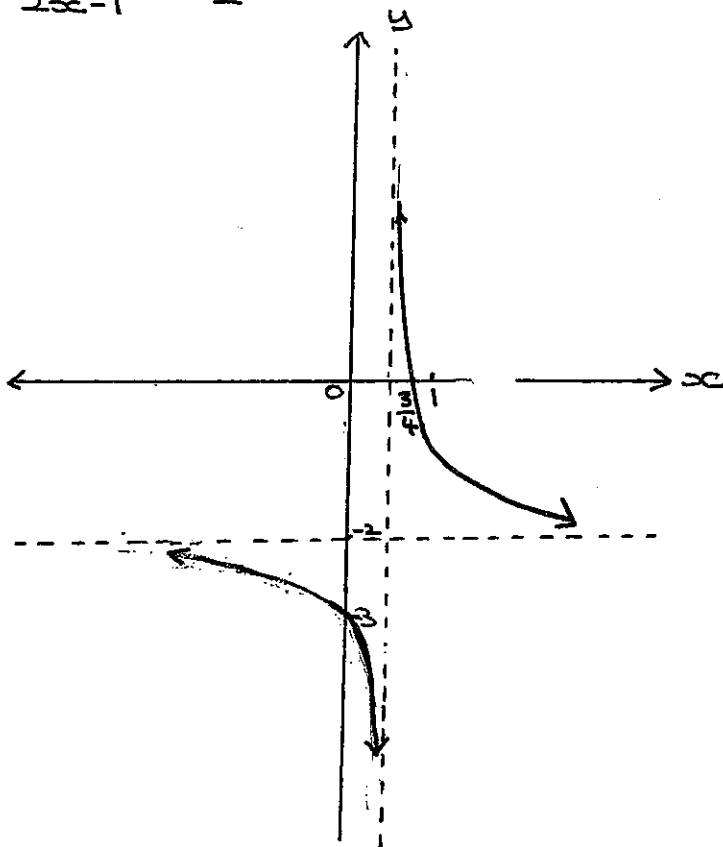
The radius is 2



b) (1) $y = (x-1)^2 + 3$



c) $y = \frac{1}{2x-1} - 2$



e) $y = \frac{1}{\sqrt{2x+9}}$

domain $2x+9 > 0$

$x > -4\frac{1}{2}$

range $y > 0$

Question 3

a) $\frac{x}{x-3} \leq 2$ NB $x \neq 3$

$$(x-3)^2 \times \frac{x}{(x-3)} \leq 2(x-3)^2$$

$$x(x-3) \leq 2(x-3)^2$$

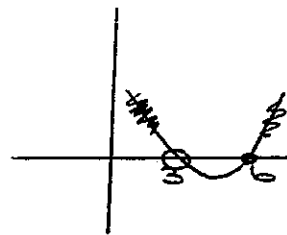
$$0 \leq 2(x-3)^2 - x(x-3)$$

$$0 \leq (x-3)[2(x-3) - x]$$

$$0 \leq (x-3)(x-6)$$

$$(x-3)(x-6) \geq 0$$

$$\therefore x < 3, x \geq 6$$



b) Let the length be x

$$2L + 2B = P$$

$$2x + 2B = 40$$

$$x + B = 20$$

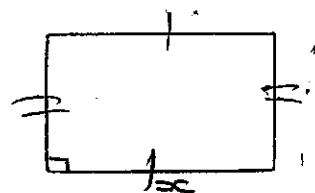
$$B = 20 - x$$

$$A = LB$$

$$A = x(20-x) \text{ as required}$$

$$2B = 40 - 2x$$

$$B = 20 - x$$



$$A = x(20-x)$$

$$96 = 20x - x^2$$

$$x^2 - 20x + 96 = 0$$

$$(x-12)(x-8) = 0$$

$$x = 8, 12$$

$\therefore$ length is 12 cm and the breadth is 8 cm

$$|x| = \pm x$$

$$|x+1| = \pm (x+1)$$

c) $||x+1|| = 2+3x$

$$x+1 = 2+3x$$

$$-2x = 1$$

$$x = -\frac{1}{2}$$

$$\underset{-3x}{(x+1)} = \underset{-3x}{2+3x}$$

$$\begin{aligned} -2x &= 1 \\ x &= -\frac{1}{2} \end{aligned}$$

or $x+1 = -(2+3x)$

$$x+1 = -2-3x$$

$$4x = -3$$

$$x = -\frac{3}{4}$$

OR $(x+1) = -(2+3x)$

Check

when $x = -\frac{1}{2}$

$$\text{L.H.S} = |x+1|$$

$$= |-\frac{1}{2}+1|$$

$$= \frac{1}{2}$$

$$\text{R.H.S} = 2+3x$$

$$= 2+3(-\frac{1}{2})$$

$$= \frac{1}{2}$$

$$\text{L.H.S} = \text{R.H.S}$$

$\therefore x = -\frac{1}{2}$ is a solution

when $x = -\frac{3}{4}$

$$\text{L.H.S} = |x+1|$$

$$= |-\frac{3}{4}+1|$$

$$= \frac{1}{4}$$

$$\text{R.H.S} = 2+3x$$

$$= 2+3(-\frac{3}{4})$$

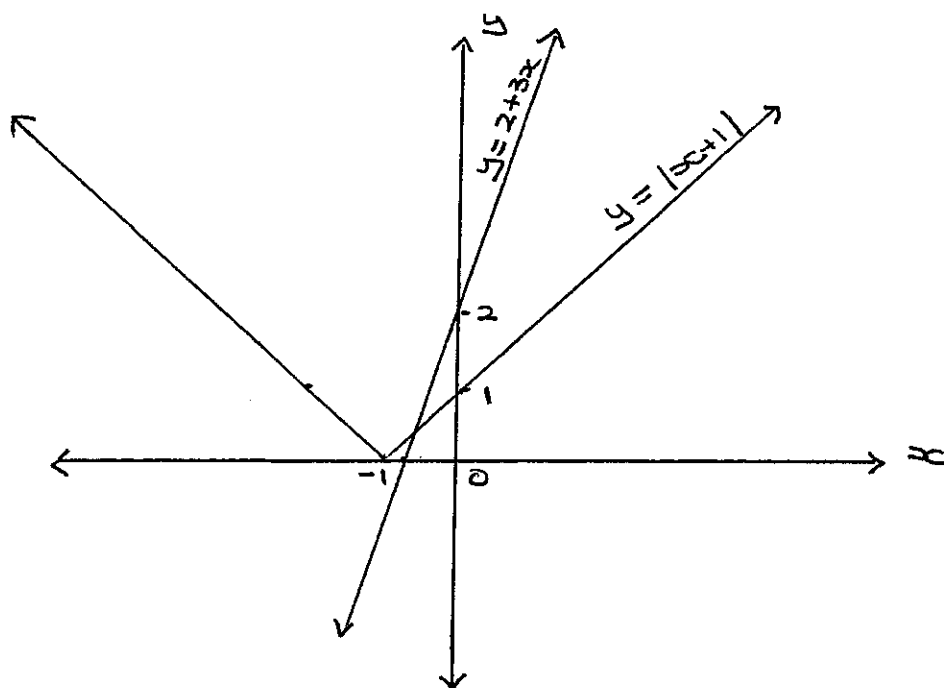
$$= -\frac{1}{4}$$

$$\text{L.H.S} \neq \text{R.H.S}$$

$\therefore x = -\frac{3}{4}$ is not a solution

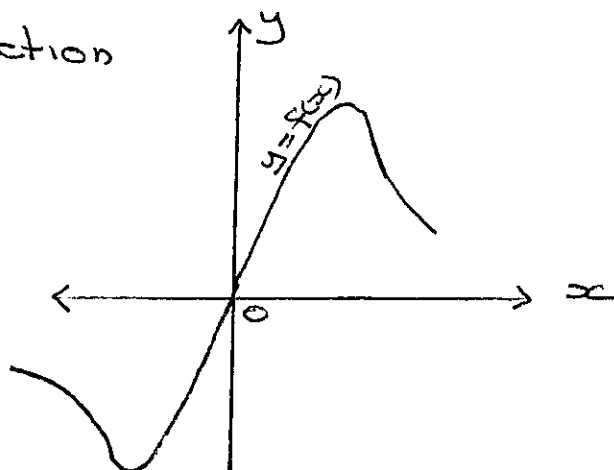
$\therefore x = -\frac{1}{2}$ is the only solution

OR

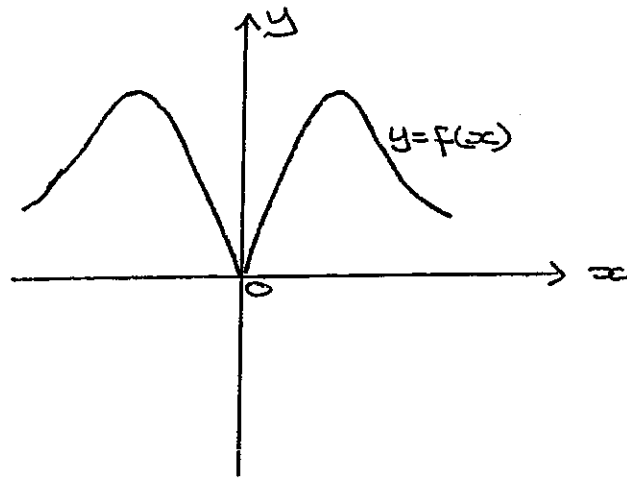


$$x = -\frac{1}{2}$$

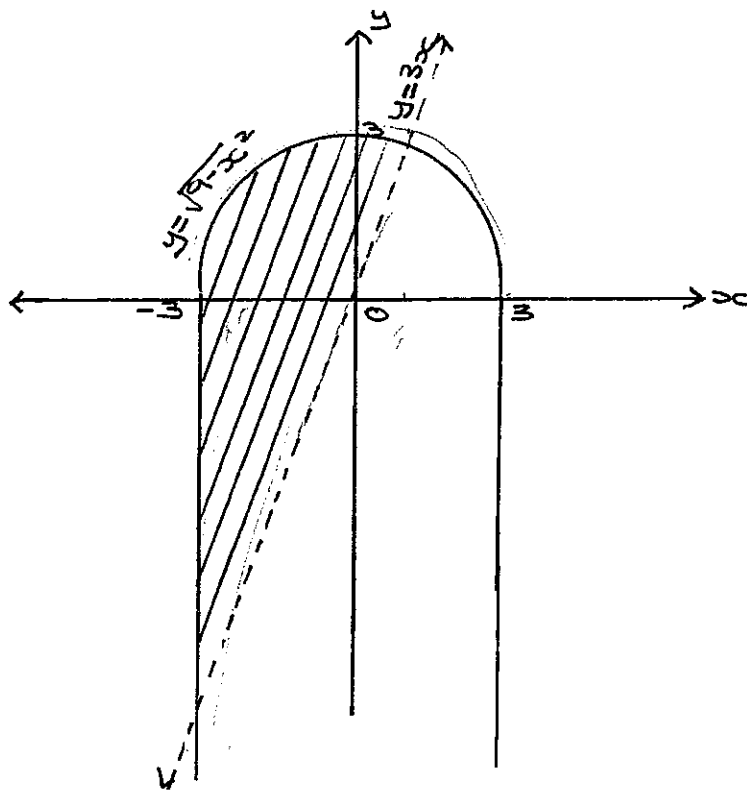
a) odd function



ii) even function



Question 4



$$\begin{aligned} \text{b) } f(x) &= 3(2^x) - 2(3^x) \\ f(x+1) &= 3(2^{x+1}) - 2(3^{x+1}) \\ &= 3 \times 2 \times 2^x - 2 \times 3^x \times 3 \\ &= 6(2^x - 3^x) \text{ as required} \end{aligned}$$

$$c) f(x) = \frac{2x^2}{x^2-16}$$

$$i) f(x) = \frac{2x}{(x-4)(x+4)}$$

$$x \neq \pm 4$$

$$ii) \lim_{x \rightarrow \infty} \frac{2x^2}{x^2-16} = \lim_{x \rightarrow \infty} \frac{\frac{2x^2}{x^2}}{\frac{x^2}{x^2} - \frac{16}{x^2}} = 2$$

$$iii) f(x) = \frac{2x^2}{x^2-16}$$

$$f(-x) = \frac{2(-x)^2}{(-x)^2-16}$$

$$= \frac{2x^2}{x^2-16}$$

$$\therefore f(x) = f(-x)$$

$\therefore f(x)$ is even

